

From Generative to Embodied Intelligence: Network Evolution for Large-AI-Model Agents

INTRO

I Section I. Introduction

Research motivations

Contributions



The diagram illustrates the progression of Large Action Models (LAMs) through three stages: Generative LAMs, Agentic LAMs, and Embodied LAMs. These stages are represented by colored boxes (light blue, light orange, and light green respectively) arranged horizontally. Below these boxes is a wide, light purple box labeled 'Key Trends in LAM Evolution'. To the left of the 'Generative LAMs' box is a red circle containing the Roman numeral 'II', indicating this is the second section of the presentation.

II Section II. From Generative to Embodied LAMs: Performance Metrics, Benchmarks, and Model Evolution

Generative LAMs Agentic LAMs Embodied LAMs

Key Trends in LAM Evolution

Generative LAMs

Agentic LAMs

Embodied LAMs

Key Trends in LAM Evolution

Section III. From Generative to Embodied Networks for LAMs: QoS Scale and Model Emphasis

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graph LR; A[Generative Stage: Fidelity and Cost under Elastic Interaction] --> B[Agentic Stage: Long-Horizon Goodput, Memory, and Retry]; B --> C[Embodied Stage: Closed-Loop Timing and Physical Loss]
```

Generative Stage: Fidelity and Cost under Elastic Interaction

Agentic Stage: Long-Horizon Goodput, Memory, and Retry

Embodied Stage: Closed-Loop Timing and Physical Loss

Generative Stage: Fidelity and Cost under Elastic Interaction

Agentic Stage: Long-Horizon Goodput, Memory, and Retry

Embodied Stage: Closed-Loop Timing and Physical Loss

IV Section IV. From Generative to Embodied Networks for LAMs: Architecture Evolution

The diagram illustrates the evolution of LAM architectures through three stages, each represented by a colored rounded rectangle:

- Architecture for Generative-LAM Networks** (Light Blue box)
- Architecture for Agentic-LAM Networks** (Light Orange box)
- Architecture for Embodied-LAM Networks** (Light Green box)

Arrows indicate a sequential flow from left to right, showing the progression from Generative to Agentic to Embodied networks.

Architecture for Generative-LAM Networks

Architecture for Agentic-LAM Networks

Architecture for Embodied-LAM Networks



v Section V. Improving Networks for LAMs: Key Enabling Technologies

```
graph TD; A[Enablers for Generative-LAM Networks] --> B[Compression and token communications]; A --> C[Split, distributed, and speculative inference];
```

Enablers for Generative-LAM Networks

- Compression and token communications**
- Split, distributed, and speculative inference**

Compression and token communications

Split, distributed, and speculative inference

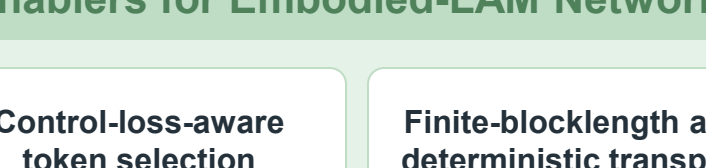
```
graph TD; A[Enablers for Agentic-LAM Networks] --> B[Reliable collaboration]; A --> C[Agentic Model and tool routing];
```

Enablers for Agentic-LAM Networks

- Reliable collaboration**
- Agentic Model and tool routing**

Reliable collaboration

Agentic Model and tool routing



Enablers for Embodied-LAM Networks

- Control-loss-aware token selection
- Finite-blocklength and deterministic transport
- Semantic and goal-oriented communication
- Communication-control co-design and dynamic model sharing

Control-loss-aware token selection

Finite-blocklength and deterministic transport

Semantic and goal-oriented communication

**Communication--control
co-design and dynamic
model sharing**

TOKEN PLANE

TRANSPORT PLANE

COORDINATION PLANE



VI Section VI. Case Study: Two-Stage Recognition of LAM Traffic

Phase I: Passive Traffic Recognition Experiment



Phase II: Active Recognition and QoS Mapping



VII Section VII. Challenges and Future Directions

Tokenizer Compatibility

Token-Native Protocols

Benchmarking, Theory, and Standardization Gaps



VIII

Section VIII. Conclusions

CLOSING